PROJECT SPECIFICATION & REQUIREMENT DOCUMENT

ENG-KG001: Semantic Knowledge Graph Construction Pipeline

Project Stream: Knowledge Graph Foundations | **Priority:** Medium | **Target Audience:** Engineering Interns | **Date:** June 2026

📜 Project Background

The engineering team has successfully completed **ENG-AI001: Legacy Dataset Profiling & Ingestion Pipeline**, where the target CSV dataset was validated, profiled, and prepared for semantic processing.

With the dataset now verified, the next stage of the pipeline is to transform traditional tabular data into a machine-readable **Knowledge Graph**.

Rather than treating data as isolated rows and columns, semantic systems organize information into **entities**, **properties**, and **relationships**. This process begins by constructing an ontology (TBox), followed by generating individual instances (ABox), which together form the final Knowledge Graph.

Your objective is to build a Python application that consumes the validated CSV output from **ENG-AI001**, constructs a semantic model, and visualizes the resulting Knowledge Graph.

🛠️ Core Engineering Directive: Responsible AI Usage

Strict Policy Regarding Generative AI Tools

- **Minimize Code Generation:** Do not use AI to write, copy-paste, or generate the implementation of your application.
- **Use AI to Learn, Not to Prompt:** Use AI only as a conceptual mentor. Ask questions such as "What is the difference between a TBox and an ABox?", "How are RDF triples represented?", or "How does RDFLib serialize Turtle files?", then implement the solution yourself.
- **The Goal:** Building a strong understanding of semantic modeling and ontology construction will prepare you for more advanced Knowledge Graph engineering and reasoning tasks.

🎯 Functional Objectives

You are required to build a modular Python application (**kg_builder.py**) that programmatically constructs a Knowledge Graph using the validated dataset generated from **ENG-AI001**. The application must execute the following semantic engineering operations:

1. **Import the AI001 Output.** Load the validated CSV dataset generated by the CSV Reader from ENG-AI001. Use the extracted rows & columns (schema) and cleaned data records as the foundation for Knowledge Graph construction.
2. **Construct the Initial TBox.** Using the extracted column headers, create an initial ontology schema. Identify classes, datatype properties, and object properties (where applicable). This represents the first draft of the Knowledge Graph schema.
3. **Refine the TBox.** Improve the initial ontology by applying semantic refinements: renaming classes, renaming properties, adding namespaces, creating meaningful relationships, and standardizing naming conventions. The refined ontology becomes the Final TBox.

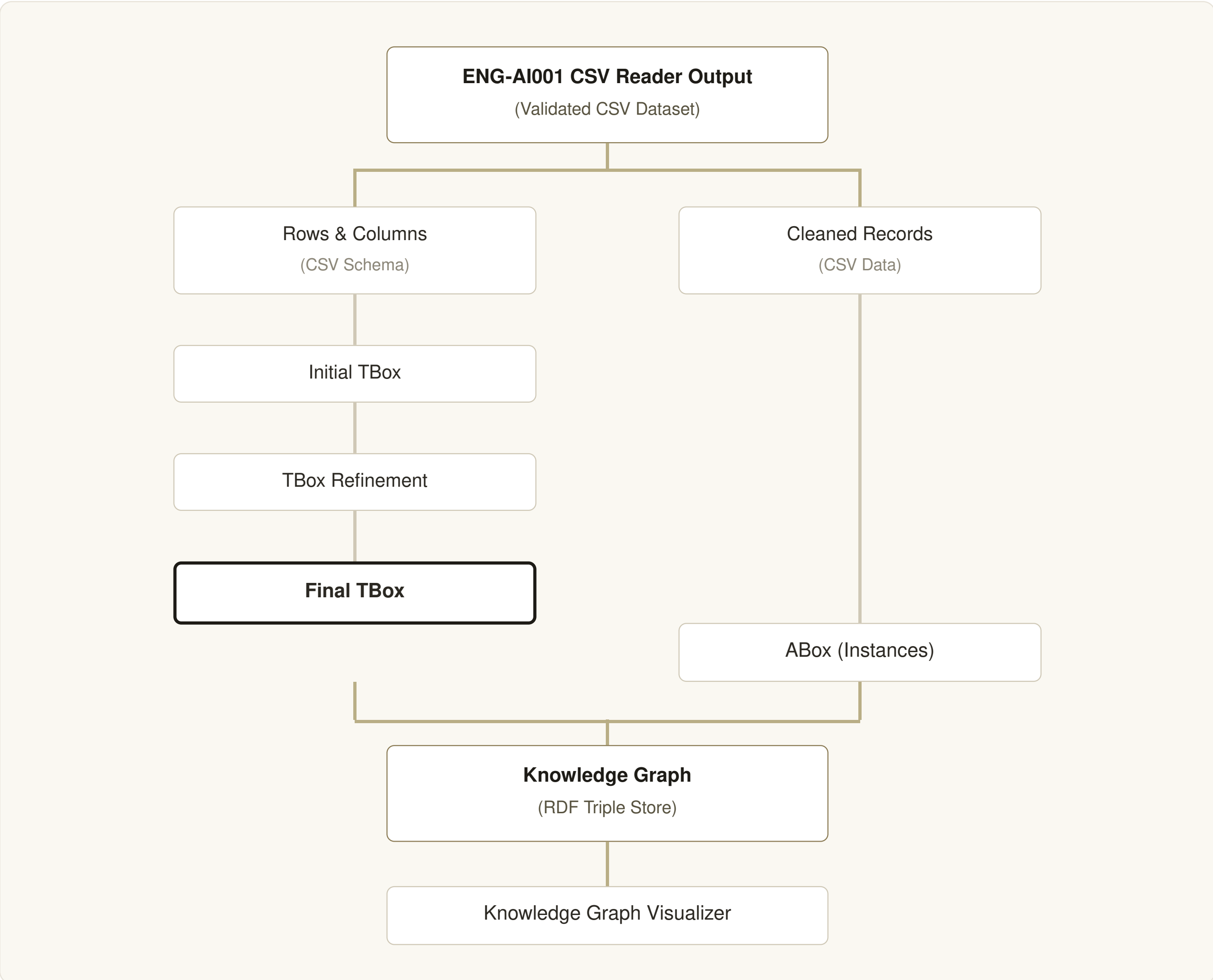
4. **Generate the ABox.** Using the cleaned data records from the AI001 CSV Reader together with the Final TBox, generate the instance layer (ABox). Each row in the dataset should become an RDF Individual with properties defined by the Final TBox.

5. **Build the Knowledge Graph.** Combine the Final TBox and generated ABox to construct the complete RDF Knowledge Graph. Serialize the graph into a Turtle file named **knowledge_graph.ttl**.

6. **Visualize the Knowledge Graph.** Generate a simple graphical visualization displaying entities (nodes), relationships (edges), and relationship labels. A simple implementation using NetworkX, Matplotlib, or PyVis is sufficient.

7. **Generate an Automated Report.** Programmatically generate a report named **kg_report.txt** summarizing total classes, total individuals, total RDF triples, total properties, and graph generation status.

🧬 Knowledge Graph Construction Pipeline



✅ Success & Acceptance Criteria

To complete this task successfully, your application and generated outputs must satisfy the following technical requirements:

- ☐ **CSV Import:** Successfully loads the validated CSV generated by ENG-AI001.
- ☐ **Ontology Construction:** Correctly generates an Initial TBox and produces a refined Final TBox.

- ☐ **Instance Generation:** Successfully creates an ABox where every CSV row becomes an RDF Individual.
- ☐ **Knowledge Graph Generation:** Successfully combines the Final TBox and ABox into a valid RDF Knowledge Graph.
- ☐ **Graph Serialization:** Automatically exports a valid knowledge_graph.ttl file.
- ☐ **Knowledge Graph Visualization:** Displays the generated Knowledge Graph with nodes and relationships.
- ☐ **Automated Report:** Automatically generates kg_report.txt summarizing the constructed Knowledge Graph.

🎁 Next-Phase Progression

Downstream Integration: Successful completion of this module unlocks **ENG-KG002: Ontology Reasoning & Semantic Validation**, where the constructed Knowledge Graph will be enhanced using ontology rules and semantic inference.

📦 Technical Submission Checklist

- ☐ **kg_builder.py** (Core Knowledge Graph construction utility)
- ☐ **knowledge_graph.ttl** (Serialized RDF Knowledge Graph)
- ☐ **kg_report.txt** (Knowledge Graph summary report)
- ☐ **README.md** (Setup instructions, dependencies, and implementation notes)
- ☐ **proof.png** (Screenshot showing successful execution and the generated Knowledge Graph visualization)

📖 Glossary of Engineering Terms

TBox (Terminological Box)
The ontology schema defining the classes, properties, and relationships used in a Knowledge Graph.

ABox (Assertional Box)
The collection of individual entities and their relationships that populate the Knowledge Graph.

Knowledge Graph
A semantic network consisting of entities connected through meaningful relationships represented as RDF triples.

RDF Triple
The fundamental building block of a Knowledge Graph consisting of: Subject → Predicate → Object.

Class
A category or type describing a group of similar entities.

Individual
A specific instance belonging to a class.

Datatype Property
A property connecting an entity to a literal value such as text or numbers.

Object Property
A property connecting one entity to another entity.

Ontology
A formal semantic model describing concepts and their relationships within a domain.

Turtle (.ttl)
A compact serialization format used to store RDF Knowledge Graphs.

Semantic Transformation
The engineering process of converting structured tabular data into an ontology-driven Knowledge Graph while preserving the meaning and relationships between data.